

The Entity–Relationship (E–R) Model

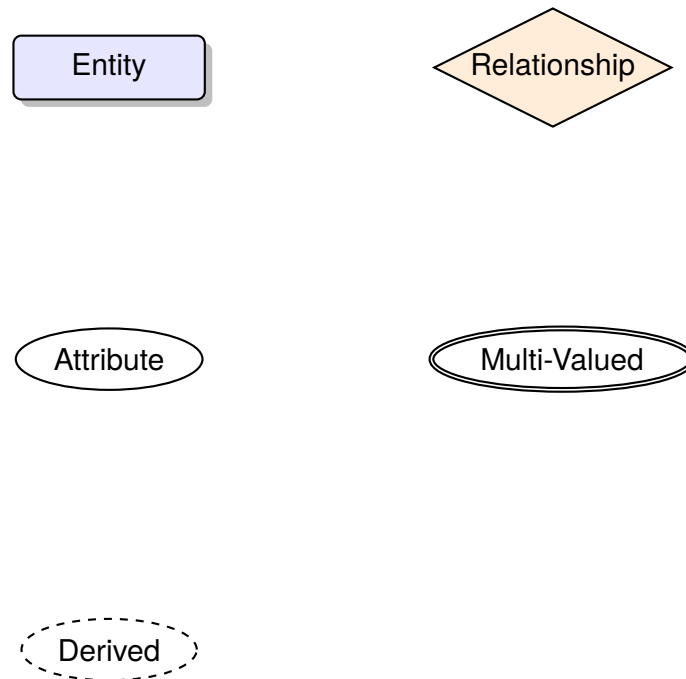
Comprehensive Notes with Examples and Diagrams

1 Introduction

The **Entity–Relationship (E–R) Model** is a high-level conceptual model for database design. It provides a graphical representation of:

- **Entities** – real-world objects or concepts.
- **Attributes** – properties of entities.
- **Relationships** – associations among entities.

2 ER Diagram Symbols Legend



3 Entity Sets

Definition

An **Entity Set** is a collection of similar entities that share the same attributes.

3.1 Strong Entity Set

- Has its own **primary key**.
- Represented by a **single rectangle**.

3.2 Weak Entity Set

- Cannot be uniquely identified by its own attributes.
- Requires a **strong entity** and a **partial key**.
- Represented by a **double rectangle**.

4 Attributes

- **Simple** – indivisible (oval).
- **Composite** – has subparts (linked ovals).
- **Single-Valued** – one value per entity (oval).
- **Multi-Valued** – multiple values allowed (double oval).
- **Derived** – computed from other attributes (dashed oval).

5 Keys

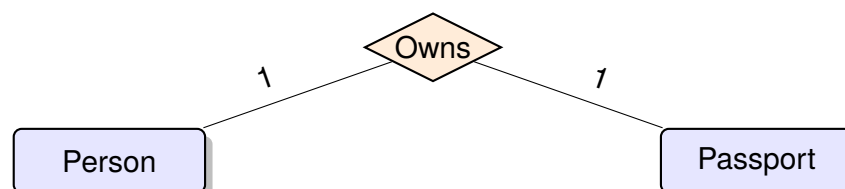
- **Primary Key** – chosen unique identifier (underlined).
- **Candidate Keys** – all possible unique identifiers.
- **Partial Key** – for weak entities; identifies entity within owner.

6 Relationships and Participation

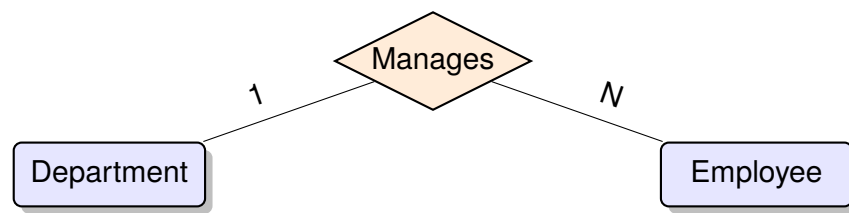
- **Total Participation** – all entities participate (double line).
- **Partial Participation** – some entities participate (single line).

7 Cardinality Constraints with Diagrams

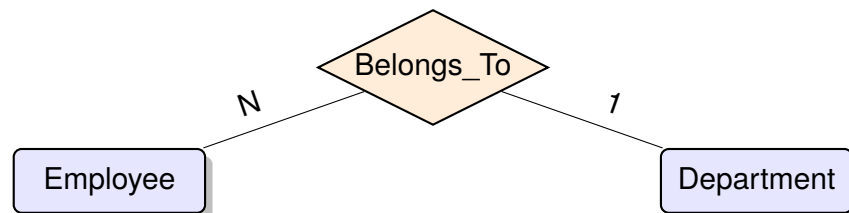
7.1 One-to-One (1:1)



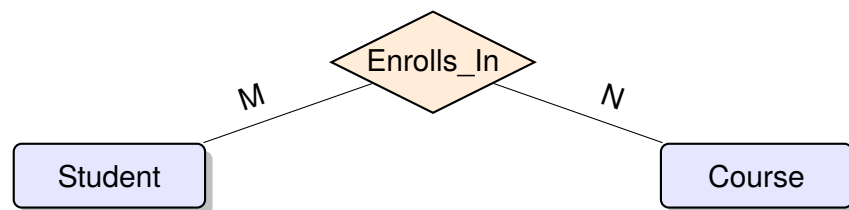
7.2 One-to-Many (1:N)



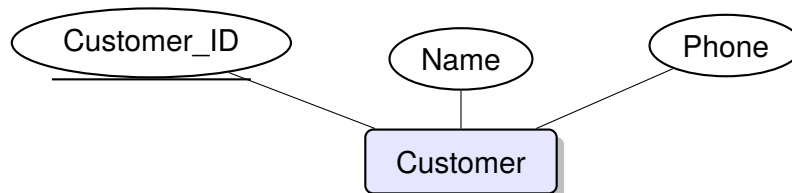
7.3 Many-to-One (N:1)



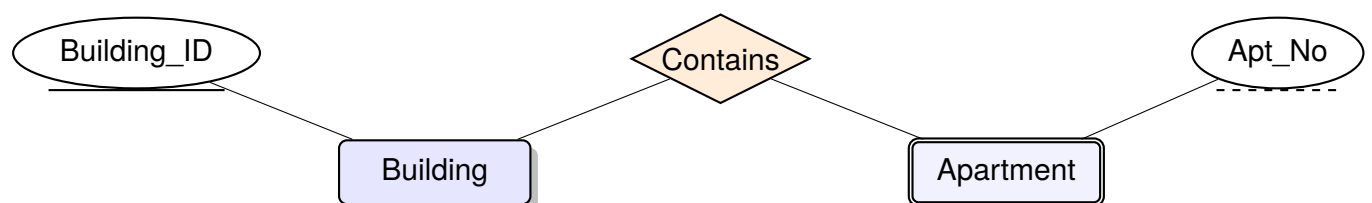
7.4 Many-to-Many (M:N)



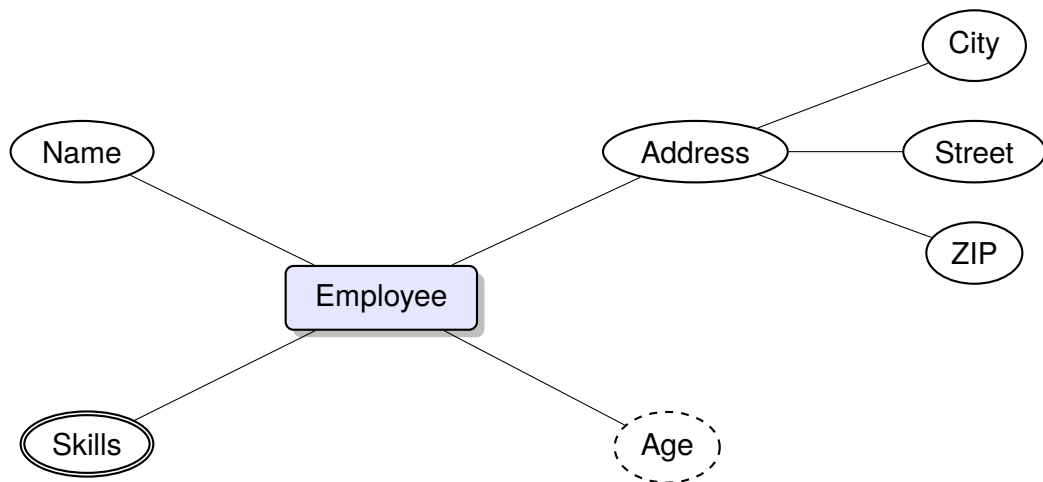
7.5 Strong Entity Set Diagram



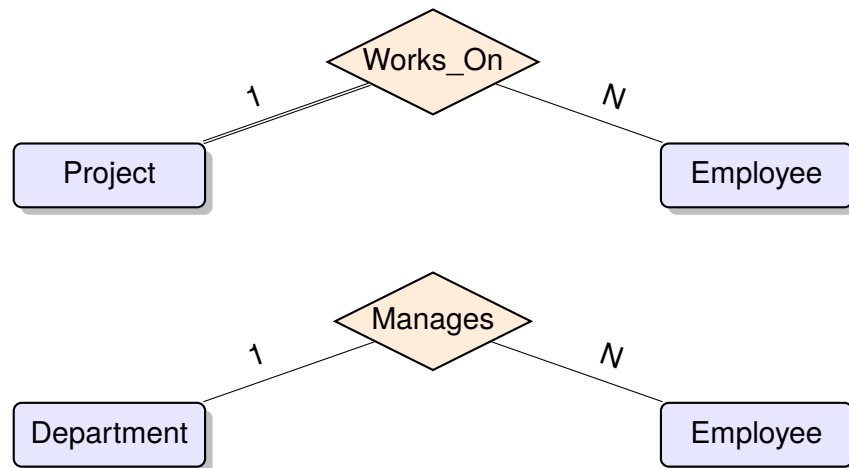
7.6 Weak Entity Set Diagram



7.7 Attributes Diagram



7.8 Participation Diagram



8 Summary Table

Concept	Description	Symbol
Strong Entity	Unique by own attributes	Rectangle
Weak Entity	Needs strong entity + partial key	Double rectangle
Simple Attribute	Indivisible property	Oval
Composite Attribute	Breakable into subparts	Linked ovals
Single-Valued Attribute	One value per entity	Oval
Multi-Valued Attribute	Multiple values allowed	Double oval
Derived Attribute	Computed from others	Dashed oval
Relationship	Association between entities	Diamond
1:1	One to one mapping	Cardinality 1:1
1:N	One to many mapping	Cardinality 1:N
N:1	Many to one mapping	Cardinality N:1
M:N	Many to many mapping	Cardinality M:N
Primary Key	Unique identifier	Underlined attribute
Partial Key	Identifier within owner	Dashed underline